

Parameters identification using experimental measurements for equivalent circuit Lithium-Ion cell models

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Abstract—Battery modelling is important for the development of efficient battery management systems (BMS), different studies on behaviour under various operating conditions, and the prediction of the batteries lifetime. The focus of this paper is on the experimental measurements on Lithium-Ion (Li-Ion) cells for the parameter identification of Equivalent Electrical Circuit Models (ECM). Depending on model accuracy, a trade-off has to be made between a simple and a complex model of the cell. A simple model is not suitable for accurately simulate the cell dynamics while a complex model is computationally expensive. Different types of equivalent circuit models are proposed in literature; first- and second-order resistor-capacitor (RC) models for Li-Ion cells are studied comparatively in this paper. Equivalent circuit parameters are identified, experimentally, and models of the cells are implemented in Matlab/Simulink. The outputs of the models are compared with actual data to calculate the error. In the models, the parameters dependency on the temperature and the age of the cell are neglected. The simulation results show that in constant current pulse discharge, the Root Mean Square Error (RMSE) expressed as a percentage for the first-order RC model is 2.78% and for the second-order RC model is 1.96%. These results are obtained without any optimization of the parameters.

Keywords—Li-Ion cell, equivalent electrical circuit models, cell modelling, parameter identification

I. INTRODUCTION

Lithium-Ion (Li-Ion) batteries belong to the most promising commercially available energy storage systems for use in many consumer electronics, such as laptops, mobile phones, digital cameras, other portable consumer electronics, renewable energy storage, and electric vehicles. As consumer awareness continues to grow and governments around the world set rigorous new fuel economy standards, the automotive sector, above all, is now on the edge of a drastic shift. This phenomenon is a strong driving force for Li-Ion battery technology, both from a scientific progress and a cost reduction point of view [1]. Due to the notable characteristics of the Li-Ion battery, including high energy density, high power, fast response, low self-discharge rate, long cycle, and calendar life, many researchers are interested and continuously developing this technology [2]. Battery performance modelling and extensive battery cell characterisation allow a shorter development time of systems or components. Accurate simulations are also a faster and safer way to study behaviour under various operating conditions. Furthermore, battery models are often needed

for real-time applications, such as in battery management systems (BMS), which observe and control Li-Ion battery cells when grouped and wired into packs. A large number of battery cell model types are investigated in the literature to predict battery behaviour, which can be divided into different perspectives (physical domains) of modelling: electrochemical models, electrical models, thermal models, mechanical models, or combinations of interdisciplinary models (e.g. electro-thermal models) [3]. The electrochemical model structure is complex to establish. It takes a lot of hardware resources, and its accuracy is not high when applied in vehicle control systems, due to the complexity of the model. Moreover, a large number of design and electrochemical kinetic parameters need to be obtained first [4], [5]. Therefore, this model is used to understand the process inside the battery to guide its designer. Electrical models, also known as equivalent electric circuit models (ECM), describe the battery operating characteristics using a circuit network. The principle is simple and the mathematical model is easy to establish and practical to use. Resistances, capacitances, and voltage sources are used to describe charging and discharging processes [6], [7], in which each component has a clear physical meaning. The first-order RC (resistor-capacitor) model was used in [8] to estimate the resistance of lithium-ion batteries and open circuit voltage and in [9] for the modelling and simulation of Li-Ion battery packs. Additionally, reference [10] used the equivalent circuit model of a second-order RC model for Li-Ion battery modelling.

Battery models must have the best trade-off between simulation accuracy, computational complexity, and parametrisation effort [1]. The higher the model complexity, the more effort there will be for testing and computing, therefore the first- and second-order RC models are studied comparatively in this paper. There are few systematic comparative studies on equivalent circuit models of lithium-ion batteries in literature [11], [12].

The complexity of the two model types is also appropriate. They are easy to implement in engineering and parameter identification with high precision is easily achieved.

This paper is organized as follows. Chapter II present the two electrical equivalent circuit models that were chosen to model the cell. The experimental setup that was used for the data acquisition is presented in chapter III. In chapter IV

the required model parameters are identified by experimental methods. Two tests are presented in detail in order to determine the data for the parameterization of the cell. From these tests, the cell capacity and cell voltage are obtained during the discharge process. It is important to state a step by step identification process for all the required parameters of the cell model. Afterwards, in the next chapter by analyze the measured data from the cell, respectively the voltage of the cell it is presented the process of determination the values of the parameters of the ECM. The simulation models are built in Matlab/Simulink environment and are presented together with the results, in chapter V. The precision of results for the two models is verified and discussed comparatively with the actual data. Finally, in the last chapter the main conclusions are summarized.

II. ELECTRICAL EQUIVALENT CIRCUIT MODELS

The equivalent electric circuit models consist of basic circuit components such as resistors, capacitors and voltage sources. First-order ECM is presented in Fig.1 and the circuit contains one resistance R_s , one network resistance, capacitor in parallel R_p, C_p and a voltage source V_{oc} . In the model, R_s denotes the ohmic resistance of the cell, R_p denote the polarization resistance and C_p denotes the polarization capacitance. The voltage source V_{oc} represent the open circuit voltage of the lithium-ion cell which varies with the cell state of charge (SOC). A state space model for the battery equivalent circuit is presented below. The states are u_1 , the voltage drop across the R_p, C_p network and u_2 the SOC.

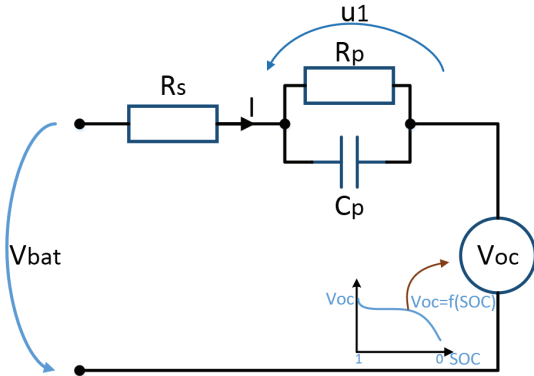


Fig. 1. First order equivalent circuit model.

$$\begin{bmatrix} \dot{u}_1 \\ \dot{u}_2 \end{bmatrix} = \begin{bmatrix} -\frac{1}{R_p C_p} & 0 \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} + \begin{bmatrix} \frac{1}{C_p} \\ \frac{1}{Q} \end{bmatrix} \cdot I \quad (1)$$

$$V_{bat} = V_{oc}(u_2) + \begin{bmatrix} 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} + R_s \cdot I \quad (2)$$

$$\frac{du_2}{dt} = \frac{I}{Q} \quad (3)$$

Where: Q is the cell capacity in Ah.

Second-order equivalent circuit contains the same components as the first-order model but a second network resistance, capacitor is added in series R_2, C_2 , Fig.2. In this configuration R_1, C_1 denote the activation polarization resistance and

capacitance and R_2, C_2 denote the concentration polarization resistance and capacitance. The state space model of the second order equivalent circuit (4,5,6):

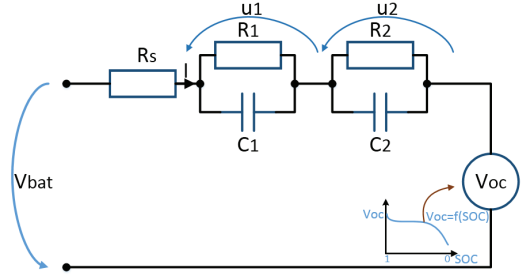


Fig. 2. Second order equivalent circuit model.

$$\begin{bmatrix} \dot{u}_1 \\ \dot{u}_2 \\ \dot{u}_3 \end{bmatrix} = \begin{bmatrix} -\frac{1}{R_1 C_1} & 0 & 0 \\ 0 & -\frac{1}{R_2 C_2} & 0 \\ 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} + \begin{bmatrix} \frac{1}{C_1} \\ \frac{1}{C_2} \\ \frac{1}{Q} \end{bmatrix} \cdot I \quad (4)$$

$$V_{bat} = V_{oc}(u_3) + \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} + R_s \cdot I \quad (5)$$

$$\frac{du_3}{dt} = \frac{I}{Q} \quad (6)$$

III. EXPERIMENTAL SETUP

After the models are described by analytical approach, it is important to express step by step the identification process for all the parameters of the battery models. For the model parameters identification, the testing bench shown in Fig.3. was established. It is composed of a NI PXIe-8135 and a programmable load (EA-EL 9400-150 0-400V 0-150A 7200W). The connection between them via analog and digital I/Os are made using a connection box. The computer is connected to the PXIe via ethernet cable and is used to control and monitor the testing bench. Data acquired is saved in the NI PXIe-1835 embedded controller memory using First-In, First-Out (FIFO) method and written in Technical Data Management Streaming (TDMS) format at 1 Hz. The software used for data acquisition and control the testing bench is LabVIEW.

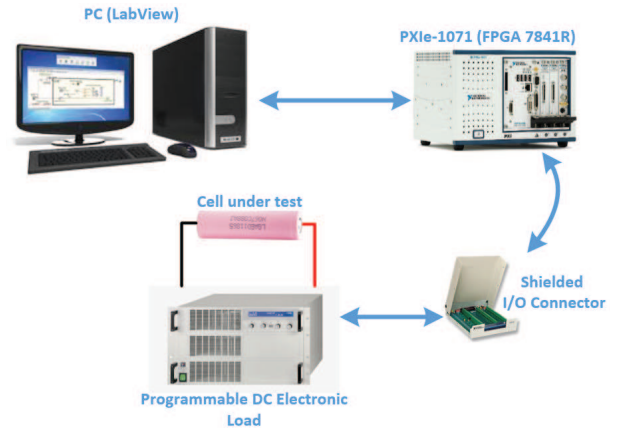


Fig. 3. Battery testing bench.

The cell used to be modeled is a Qulit Fire 18650 Li-Ion rechargeable cell with a rated capacity of 4500mAh, 3.7V rated, 4.2V maximum over charge voltage, 2.7V minimum discharge voltage.

IV. CHARACTERIZATION PROCEDURE

Before the characterization procedure the cell was charged using first constant current (CC) at 1A and after constant voltage (CV) charging method until the current drained from the charging source is less than 0.02A. After complete charging the cell is left in open circuit for one hour to reach electrochemical equilibrium. The cell was complete discharge in the tests but in many application the cell/battery is discharged until 5-10%. The complete discharge of the Lithium cells will dramatically shorten the life cycle thus a limit for the Depth of Discharge (DoD) is imposed to prolong the life cycle.

A. Capacity test

The cell capacity represents the maximum amount of energy that can be extracted from the cell under certain specified conditions. The actual energy storage capabilities of the cell can vary significantly from the "nominal" rated capacity, as the cell capacity depends strongly on the age and history of the cell, the charging or discharging regimes of the cell and the temperature. A nominal capacity value is always provided by the manufacturer inside the cell datasheet however, a test should be performed to calculate a more reasonable value, the actual capacity of the cell, to be employed inside the cell model. The cell capacity printed by the manufacturers is obtained by discharging the cell at the maximum constant current that a cell can supply for 20 hours at 20C. Several discharging tests were performed on the cell, therefore a constant current discharge at C/9 (0.5A), C/4.5 (1A), C/2.25 (2A), C/1.5 (3A) and 0.9C (4A) are applied, until the lower voltage threshold (2.7 V) is reached. The voltages waveforms are depicted in Fig. 4. The value of the capacity used in the model is the average of the capacity obtained for different constant discharging currents, 2.08Ah. From the capacity measurements it can be observed that is a big difference between the capacity obtained at 0.5A discharging constant current, 2.25Ah, and the capacity given by the manufacturer, 4.5Ah.

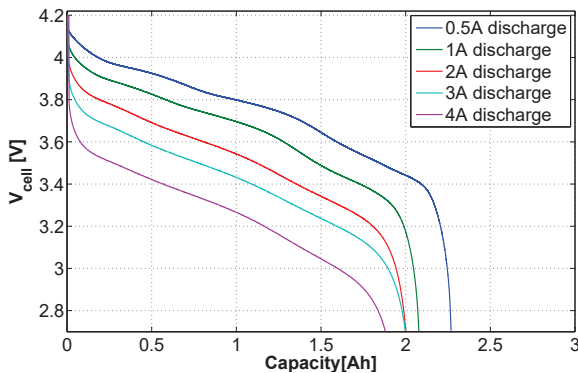


Fig. 4. Cell capacity for different values of discharge constant current.

B. Current pulse test

This test permits the proper characterization of the ECM by requiring the battery with discharging current pulses while

observing its voltage response. In order to fully parameterize the equivalent circuit elements, pulses must be performed over the whole battery range of the SOC.

The test procedure implemented in this work in order to obtain the relationship between the battery OCV and state of charge (SOC) consists of fully charge the battery with the standard charging method and afterwards series of 2A discharging current pulses from 100% to 0% state of charge (SOC), for 2 minutes separated by 8-minute rest period in between until the lower voltage threshold is reached (2.7 V) are applied. For discharging the cell from 100% to 0%, 32 discharging pulses were applied, one pulse discharge the cell approximately 3%.

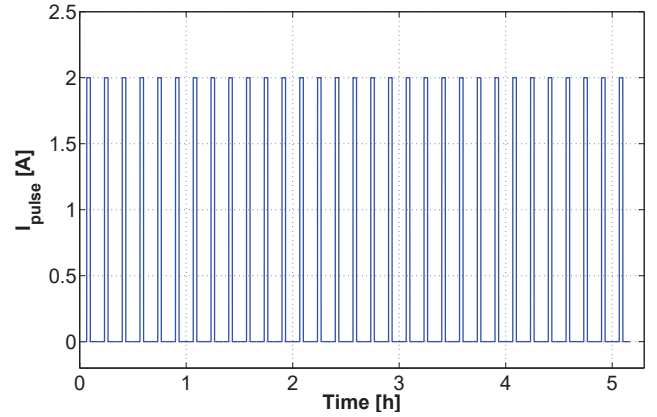


Fig. 5. The discharging current pulses.

The open circuit voltage (OCV) of the battery represents the value of the voltage when the battery is left in the open circuit condition after a discharge pulse. Therefore, after discharge pulse, the battery terminal voltage will gradually rise to a steady value after the load is removed. The OCV extracted values are plotted in Fig. 6.

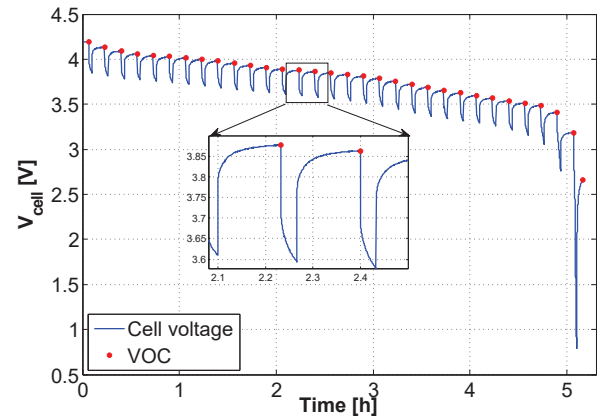


Fig. 6. The battery voltage and the values for the open circuit voltage.

C. Parameter extraction

The acquisitioned voltage is represented in Fig.6. and correspond to a complete cycle of discharge with 2A current pulses. The values of the parameters for both models are obtained from the relaxation periods following to the current pulses

when the SOC of the cell is at 60%. Fig.7. below shows a closer look at a single pulse. This pulse provides useful information about the OCV and the circuit dynamics at the given SOC.

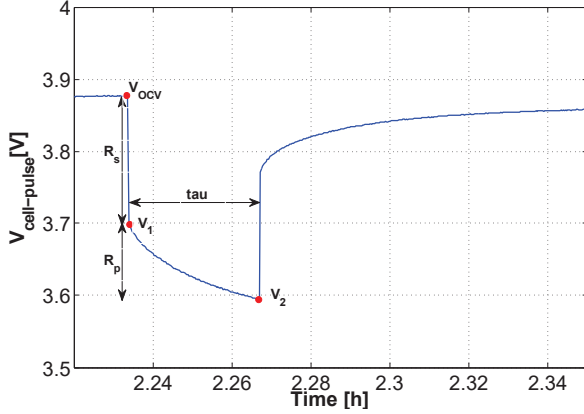


Fig. 7. Cell voltage for one discharge pulse (first-order model).

The first parameter that is needed, the OCV, is obtained as the last voltage value at the end of the rest period and is saved in an lookup table depending on the cell state of charge. The second parameter, R_s , the ohmic resistance results from the sudden voltage drop when the current pulse is applied. R_s can be calculated by (7) where I_{bat} is the amplitude of the current step.

$$R_s = \frac{V_{ocv} - V_1}{I_{bat}} \quad (7)$$

$$R_p = \frac{V_1 - V_2}{I_{bat}} \quad (8)$$

The polarization resistance R_p is responsible for the slower variation of the voltage, when the discharge current pulse is applied, after the sudden voltage drop R_s .

The polarization capacitance C_p , is calculated in 9.

$$\tau = R_p \cdot C_p \quad (9)$$

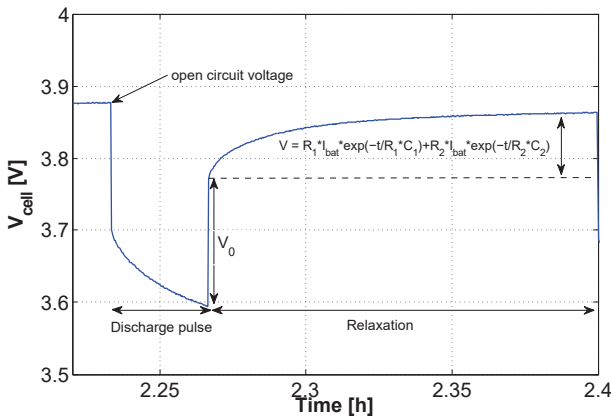


Fig. 8. Cell voltage for one discharge pulse (second-order model).

For the second order ECM in addition, other parameters are needed, that can be determined from measured voltage data using an exponential-function fitting method. The transient

response of terminal voltage data at 60% SOC was utilized to calculate these parameters.

According to Fig. 8 the terminal voltage during relaxation period can be fitted with the exponential function detailed in equation 10.

$$V_{bat} = k_0 - k_1 \cdot \exp(-a \cdot t) - k_2 \cdot \exp(-b \cdot t) \quad (10)$$

At the same time the terminal voltage in relaxation can be expressed as 11.

$$V_{bat} = V_{OC} - R_s \cdot I_{bat} - R_1 \cdot I_{bat} \cdot \exp(-t/R_1 \cdot C_1) - R_2 \cdot I_{bat} \cdot \exp(-t/R_2 \cdot C_2) \quad (11)$$

From 10 and 11 the values for R_1 , C_1 , R_2 , C_2 were obtained.

$$\begin{aligned} R_1 &= k_1 / I_{bat} \\ R_2 &= k_2 / I_{bat} \\ C_1 &= 1/a \cdot R_1 \\ C_2 &= 1/b \cdot R_2 \end{aligned} \quad (12)$$

In this paper, the curve fitting function *fit* from Matlab environment was used. In this function the Nonlinear Least Square method was chosen and in *fittype* the equation 10 was implemented. After the fitting process the values for the coefficients a , b , k_1 , k_2 are saved in a matrix in order to calculate the required parameters using the equations 12. Finally, the obtained parameters are introduced in the model simulation.

TABLE I
FIRST AND SECOND ORDER CELL MODEL PARAMETERS

Parameter	Value	Unit
R_s	0.09825	Ω
R_p	0.0519	Ω
C_p	1926.8	F
R_1	0.003	Ω
C_1	690.349	F
R_2	0.0606	Ω
C_2	1700.084	F

V. MODEL SIMULATION AND RESULTS

The state space equations presented in chapter two for the first- and second-order cell model are implemented in Matlab/Simulink, Fig.10. Therefore, the model validation is done by comparing the measured waveforms with the simulated ones during the discharge current pulse test. For these models and for the real cell the same discharge pulses were applied. The inputs for the Simulink models are the current applied, cell internal resistance, R and C network parameters, open circuit voltage, and the outputs are the cell voltage, state of charge and the voltage across the RC network. The open circuit voltage was saved into a lookup table depending on the cell state of charge, Fig.9. In the model the SOC is calculated by integrating the current divided by the cell capacity in Ah, this method is called Coulomb Counting. For a same type of cell, if the SOC is unknown, it can be obtained by measuring the cell voltage and used as input for the lookup table where the SOC is represented depending on the open circuit voltage, but the accuracy is around 10-15% because the cell temperature was not been taken into account.

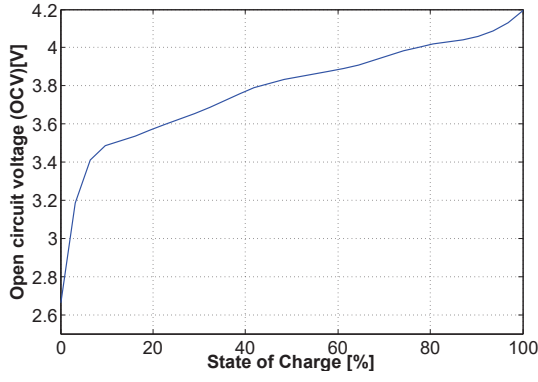


Fig. 9. Open circuit voltage

The voltages for the measured cell, for the first-order model and second-order model cell are presented in Fig.11.

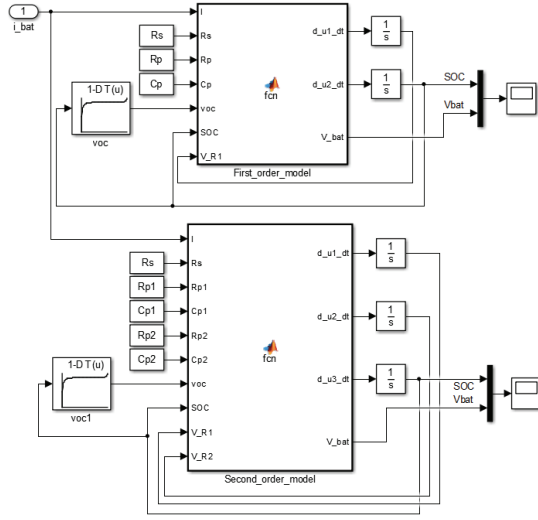


Fig. 10. Simulink models for the cells

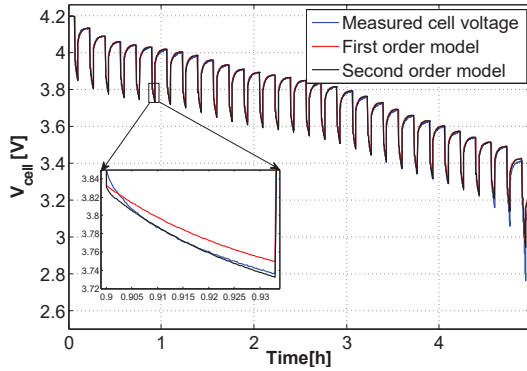


Fig. 11. Voltages for the real and modeled cells.

It is observable that the voltage for the second order model is closer to the measured cell voltage. The absolute error for the first and second order models are depicted in Fig.13 . The second order model have a smaller error.

Another important aspect is that the model behaves better during the relaxation period since the equivalent circuit parameters have been extracted during this phase.

The Root Mean Square Error (RMSE) for both models are calculated using equation 13. This error is calculated because is a better indicator of the overall model accuracy. For the first order model the calculated global RMSE is 0.1171V and for the second order model the error is 0.0825V.

$$RMSE = \sqrt{\frac{1}{T} \cdot \int_0^T [V_{\text{measured}}(t) - V_{\text{model}}(t)]^2 dt} \quad (13)$$

where T is the time period.

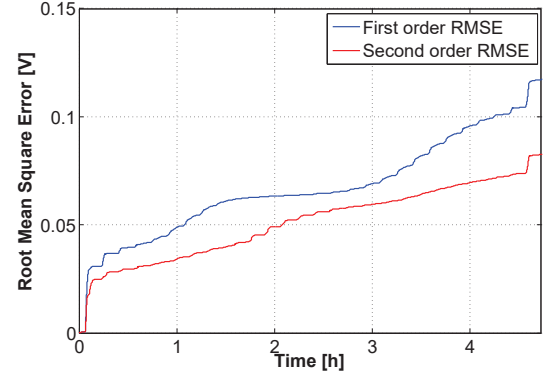


Fig. 12. Root mean square error .

By applying the discharging current pulses the calculated state of charge of the cell obtained is represented in Fig.14.

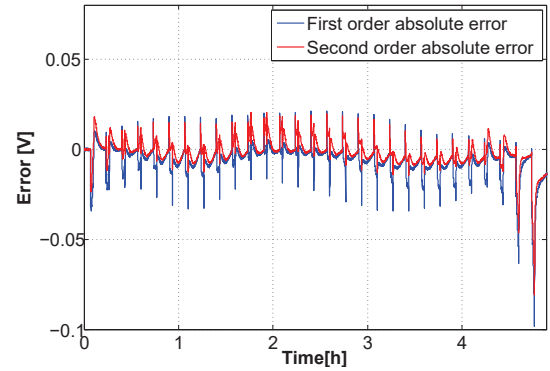


Fig. 13. Absolute error.

The depth of discharge was set at 5% in order to increase the cell life cycle.

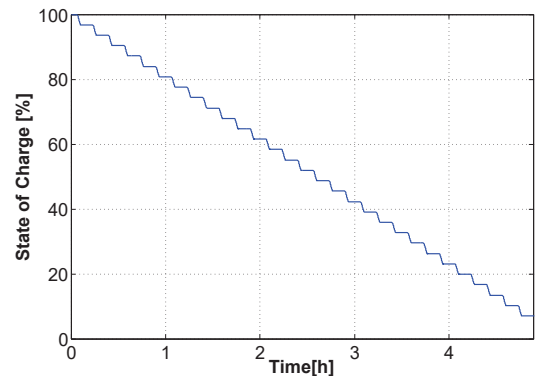


Fig. 14. Cell state of charge.

VI. CONCLUSION

In this paper, the presented parameter extraction of the parameters of a real cell and the model validation with the real data that was obtained using two equivalent circuit models aim to compare both models in order to validate the performance and the dynamical precision. It has been shown that the nominal capacity given by the datasheet was not the same as the resulted value that was obtained during the capacity test, therefore for the Matlab/Simulink model, the real value was used. The proposed extraction method and calculating procedure can be applied for extracting the parameters for any types of Lithium-Ion batteries. The results showed that the second order ECM has a high level of fidelity in relation to the measured data, having an RMSE of 0.0825V, a little smaller than the error of the first order ECM. Because the parameter extraction presented was done only at 60% SOC future work will focus on extraction of the parameters during the entire SOC range.

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